

LegalEase

Backend API with FastAPI

Detailed Project Topic Document

Purpose: Implement the backend contract between the Streamlit frontend and the AI/document services.

The project documentation specifies FastAPI, a root endpoint, modular routes, a Pydantic DocumentRequest model, and a POST /generate endpoint. ■filecite■turn0file0■L192-L207■

Pydantic request model

```
from pydantic import BaseModel, Field

class DocumentRequest(BaseModel):
    document_type: str = Field(min_length=2, max_length=120)
    parties: str = Field(min_length=2, max_length=3000)
    terms: list[str] = Field(default_factory=list, max_length=30)
    effective_date: str = Field(min_length=2, max_length=100)
    jurisdiction: str = ""
    additional_instructions: str = ""
```

FastAPI application

```
from fastapi import FastAPI
from fastapi.middleware.cors import CORSMiddleware
from .routes import router

app = FastAPI(
    title="LegalEase",
    version="1.0.0",
)

app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],
    allow_methods=["GET", "POST"],
    allow_headers=["*"],
)

app.include_router(router)

@app.get("/health")
def health():
    return {"status": "ok"}
```

Generation endpoint

```
@router.post("/generate")
def generate_document(
    request: DocumentRequest,
    generator: GeminiDocumentGenerator = Depends(get_generator),
):
    try:
        content = generator.generate_document(
            document_type=request.document_type,
            parties=request.parties,
            terms=request.terms,
            effective_date=request.effective_date,
            jurisdiction=request.jurisdiction,
            additional_instructions=request.additional_instructions,
        )
    except RuntimeError as exc:
        raise HTTPException(status_code=503, detail=str(exc))

    return {
        "document_type": request.document_type,
        "content": content,
    }
```

Export endpoints

```
POST /export/txt
POST /export/docx
POST /export/pdf
```

These endpoints keep document formatting on the backend. The frontend only sends the final edited text and receives a downloadable binary response.

Source basis: This topic is derived from the supplied LegalEase project documentation. Where the source did not provide implementation detail, this document presents the corresponding implementation structure used in the completed project.